

Mathematics A

Unit 2 • Chapter OL-T20 • Expertise Q16+ Classified

Cross-session • chapter-organised candidate

2 classified items • 3 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

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ELITE TEACHING EDITION - REVIEWED FOR WEBSITE USE

Contents

Unit 2 • Expertise • Unit 2 • Chapter OL-T20 • Expertise Q16+ • 2 items • 3 marks

Chapter	Items	Marks	Starts
01. OL-T20 Completing the Square	2	3	4

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Completing the Square

Unit 2 • Expertise • 3 marks • 2 item(s)

November 2021 | 4MA1-1H Q19 | 2 marks

Negative leading coefficient or maximum form

November 2021 | 4MA1-1H Q19 | 1 marks

Negative leading coefficient or maximum form

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

19 $ABCED$ is a five-sided shape.

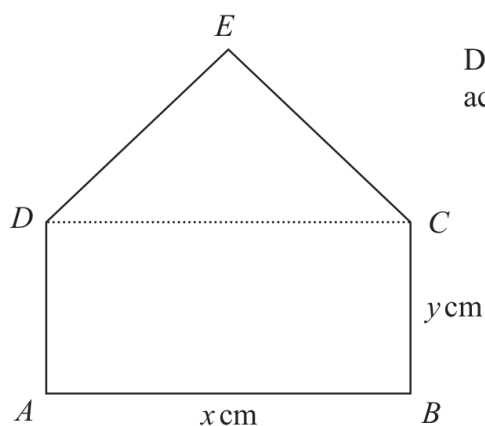


Diagram **NOT**
accurately drawn

$ABCD$ is a rectangle.

CED is an equilateral triangle.

$$AB = x \text{ cm} \quad BC = y \text{ cm}$$

The perimeter of $ABCED$ is 100 cm.

The area of $ABCED$ is $R \text{ cm}^2$

(a) Show that $R = \frac{x}{4} \left(200 - [6 - \sqrt{3}]x \right)$

(3)

(b) (i) Find the value of x for which R has its maximum value.

Give your answer in the form $\frac{p}{q - \sqrt{3}}$ where p and q are integers.

$$x = \dots\dots\dots$$

(2)

(ii) Explain why the maximum value of R is given by this value of x .

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(1)

YOUR WORKING

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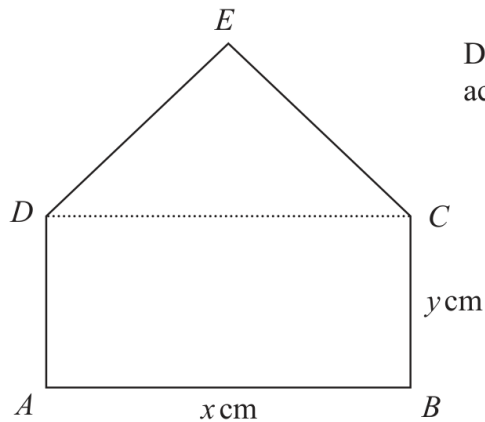


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